

Linked List – One Page Memory Map + Code (Interview Ready)

CORE STRUCTURE

```
class Node {
  constructor(data, next = null) {
    this.data = data;
    this.next = next;
  }
}

class LinkedList {
  constructor() {
    this.head = null;
  }
}
```

INSERT OPERATIONS

```
// Insert at Beginning
insertAtBeginning(data) {
  this.head = new Node(data, this.head);
}

// Insert at End
insertAtLast(data) {
  const newNode = new Node(data);
  if (!this.head) {
    this.head = newNode;
    return;
  }
  let curr = this.head;
  while (curr.next) curr = curr.next;
  curr.next = newNode;
}
```

DELETE OPERATIONS

```
// Delete First
deleteFirstNode() {
  if (!this.head) return;
  this.head = this.head.next;
}

// Delete Last
removeLastNode() {
  if (!this.head) return;
  if (!this.head.next) {
    this.head = null;
    return;
  }
  let curr = this.head;
  while (curr.next.next) curr = curr.next;
  curr.next = null;
}

// Delete by Value
removeInBetweenElement(key) {
  if (!this.head) return;
  if (this.head.data === key) {
    this.head = this.head.next;
    return;
  }
  let curr = this.head;
```

```

while (curr.next) {
  if (curr.next.data === key) {
    curr.next = curr.next.next;
    return;
  }
  curr = curr.next;
}
}

```

TRAVERSAL & SEARCH

```

printList() {
  let curr = this.head;
  let res = [];
  while (curr) {
    res.push(curr.data);
    curr = curr.next;
  }
  console.log(res.join(" -> "));
}

```

```

search(key) {
  let curr = this.head;
  while (curr) {
    if (curr.data === key) return true;
    curr = curr.next;
  }
  return false;
}

```

REVERSE & CYCLE DETECTION

```

// Reverse Linked List
reverseList() {
  let prev = null;
  let curr = this.head;
  while (curr) {
    let next = curr.next;
    curr.next = prev;
    prev = curr;
    curr = next;
  }
  this.head = prev;
}

```

```

// Detect Cycle (Floyd)
detectCycle() {
  let slow = this.head;
  let fast = this.head;
  while (fast && fast.next) {
    slow = slow.next;
    fast = fast.next.next;
    if (slow === fast) return true;
  }
  return false;
}

```